Investigating The Rate of a Reaction

Exam code: 0620 & 0971

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Investigating the rate of a reaction

- To measure the **rate of a reaction**, we need to be able to measure:
  - How quickly the reactants are used up
    OR
    How quickly the products are formed
- The method used for measuring depends on the substances involved
- There are a number of ways to measure a reaction rate in the lab
  - They all depend on a property changing during the course of the reaction
  - Properties that change during the course of a reaction include:
    - Colour
    - Mass
    - Volume
- The changing property is taken to be **proportional** to the concentration of the reactant or product
- Faster reactions can be easier to measure when the reaction is over
  - This can be done by averaging a collected measurement over the course of the reaction
- Some reaction rates can be measured as the reaction proceeds (this generates more data)
- Three commonly used techniques are:
  - **measuring mass loss on a balance**
  - **measuring the volume of a gas produced**
  - **measuring a reaction where there is a colour change at the end of the reaction**

Investigating the effect of concentration of a solution on the rate of reaction

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

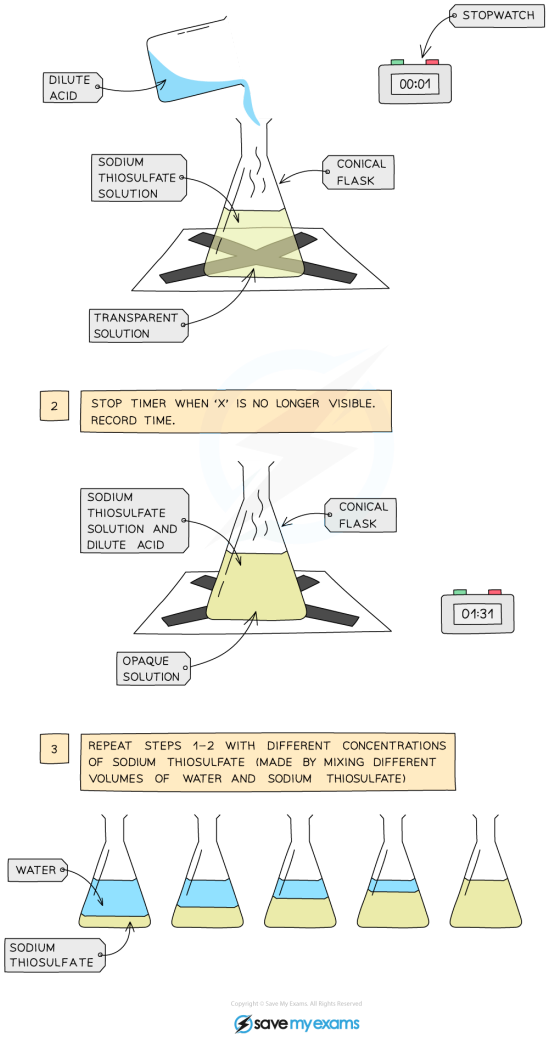

Diagram showing the apparatus needed to investigate the effect of concentration on the rate of reaction

Method:

- Measure 50 cm^3 of sodium thiosulfate solution into a flask
- Measure 5 cm^3 of dilute hydrochloric acid into a measuring cylinder
- Draw a cross on a piece of paper and put it underneath the flask
- Add the acid into the flask and immediately start the stopwatch
- Look down at the cross from above and stop the stopwatch when the cross can no longer be seen
- Repeat using different concentrations of sodium thiosulfate solution (mix different volumes of sodium thiosulfate solution with water to dilute it)

Result:

- With an increase in the concentration of a solution, the rate of reaction will increase
- This is because there will be more reactant particles in a given volume, allowing more frequent and successful collisions, increasing the rate of reaction

Help me understand

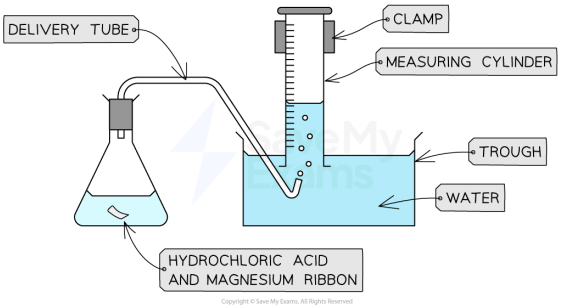

Diagram showing the process of downwards displacement to investigate the effect of the surface area of a solid on the rate of reaction

Method:

- Add dilute hydrochloric acid into a conical flask
- Use a delivery tube to connect this flask to a measuring cylinder upside down in a bucket of water (downwards displacement)
- Add magnesium ribbon to the conical flask and quickly put the bung back into the flask
- Measure the volume of gas produced in a fixed time using the measuring cylinder
- Repeat with different size pieces of magnesium ribbon
    - The same total mass of magnesium must be used

Result:

- Smaller pieces of magnesium ribbon cause an increase in the surface area of the solid, so the rate of reaction will increase
- This is because more surface area of the particles will be exposed to the other reactant so there will be more frequent and successful collisions, increasing the rate of reaction

Investigating the effect of temperature on the rate of reaction

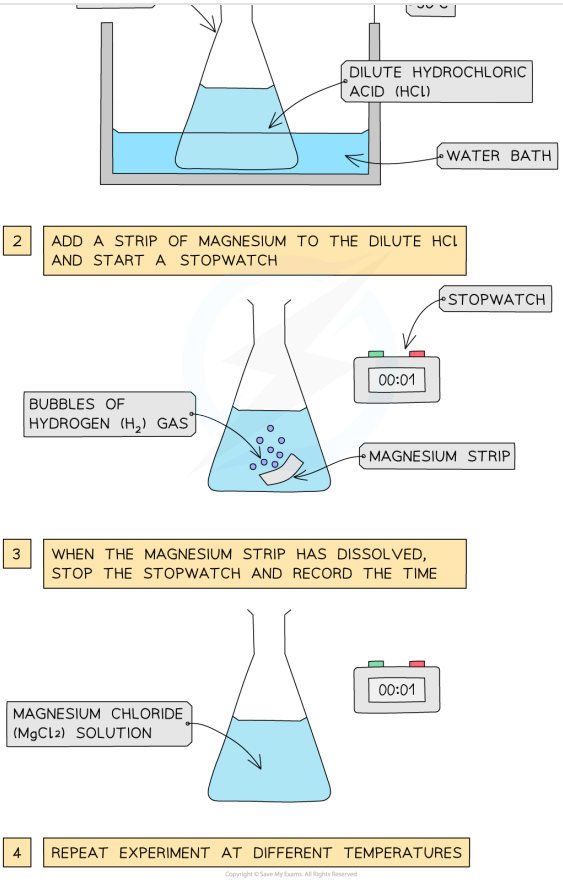

Diagram showing the apparatus needed to investigate the effect of temperature on the rate of reaction

Method:

- Dilute hydrochloric acid is heated to a set temperature using a water bath
- Add the dilute hydrochloric acid into a conical flask
- Add a strip of magnesium and start the stopwatch
- Stop the time when the magnesium fully reacts and disappears
- Repeat at different temperatures and compare results

Result:

- With an increase in the temperature, the rate of reaction will increase
- This is because the particles will have more kinetic energy than the required activation energy, therefore more frequent and successful collisions will occur, increasing the rate of reaction

Investigating the effect of a catalyst on the rate of reaction

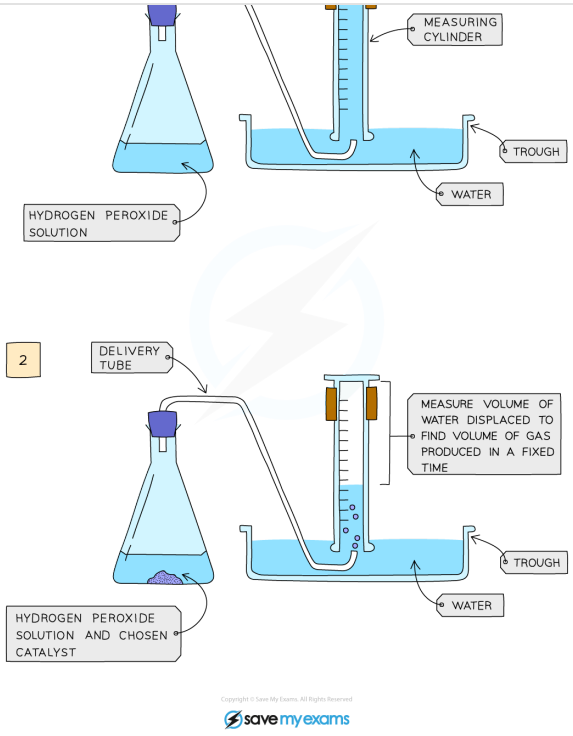

Diagram showing the apparatus needed to investigate the effect of a catalyst on the rate of reaction

Method:

- Add hydrogen peroxide into a conical flask
- Use a delivery tube to connect this flask to a measuring cylinder upside down in a tub of water (downwards displacement)
- Add the catalyst manganese(IV) oxide into the conical flask and quickly place the bung into the flask
- Measure the volume of gas produced in a fixed time using the measuring cylinder
- Repeat experiment without the catalyst of manganese(IV) oxide and compare results

Result:

- Using a catalyst will increase the rate of reaction
- The catalyst will provide an alternative pathway requiring lower activation energy so more colliding particles will have the necessary activation energy to react
- This will allow more frequent and successful collisions, increasing the rate of reaction

Monitoring changes in mass

- Many reactions involve the production of a gas which will be released during the reaction
- The gas can be collected and the volume of gas monitored as per some methods above
- Alternatively, the reaction can be performed in an **open flask** on a balance to measure the loss in mass of reactant
- Cotton wool is usually placed in the mouth of the flask which allows gas out but prevents any materials from being ejected from the flask (if the reaction is vigorous)

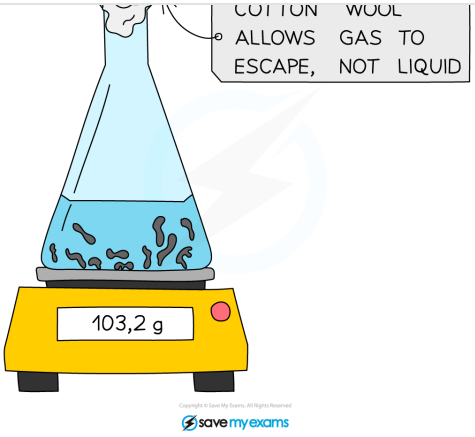

Diagram showing the set-up for measuring the rate of reaction by loss in mass

- This method is not suitable for hydrogen and other gases with a small relative formula mass, M_r as the loss in mass may be too small to measure

Examiner Tips and Tricks

There are many different methods of investigating the rate of reaction.

Another method of gas collection you may see uses a gas syringe.

Extended tier students may be required to devise and evaluate methods of investigating rates of reaction.

Evaluating investigations of rates of reactions

Extended tier only

- When investigating rates of reaction, there are a number of different methods that can be used to carry out the same investigation
- Evaluating what is the best method to use is part of good experimental planning and design
- This means appreciating some of the advantages and disadvantages of the methods available

Advantages and disadvantages of methods of investigating rates of reaction

Experiment	Advantage	Disadvantage
Formation of a solid /	Simple experiment	Difficult to determine when the cross is

cross experiment)		different levels of cloudiness Easy to contaminate equipment
Gas collection using a gas syringe	Works for all reactions that produce a gas All the gas collected is from the reaction Easy to set up	Gas syringes are fragile and expensive Gas syringes can stick They can collect limited volumes Gas can be lost before the bung is securely connected to the reaction flask, especially in fast reactions
Gas collection using an inverted measuring cylinder	Works for all reactions that produce a gas Uses common lab equipment	The delivery tube can pop out of the measuring cylinder It can be difficult to read the scale as it is upside down and may be obscured by bubbles Gas is lost while the bung is connected to the reaction flask
Measuring mass lost on a balance	Easy to set up Uses common lab equipment	Not suitable for gases with low molecular mass

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